Bayes Filter and Kalman Filter

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***Abstract*—Autonomous vehicles require accurate state estimation despite uncertainties in sensor measurements and vehicle motion. The Bayes Filter combines motion predictions with sensor observations to estimate the vehicle's state. The Kalman Filter simplifies this process under linear and Gaussian assumptions, providing an efficient method for state estimation and sensor fusion. This paper discusses the Bayes Filter, the assumptions and equations of the Kalman Filter, and its applications in autonomous vehicles.**

Keywords— Bayes Filter, Kalman Filter, State Estimation, Sensor Fusion, Autonomous Vehicles

I. Introduction

An autonomous vehicle must continuously estimate its position, velocity and orientation to navigate safely. However, sensor measurements are often affected by noise, and vehicle motion models are not perfectly accurate. Therefore, relying on a single measurement may lead to inaccurate estimates.

Filtering techniques address this problem by combining information from the vehicle's motion model and sensor measurements. The Bayes Filter provides a general probabilistic framework for this process, while the Kalman Filter offers an efficient solution under specific assumptions.

II. Bayes Filter

The Bayes Filter is a probabilistic method used to estimate the state of a system when its exact state is unknown. It combines information from a motion model, which predicts how the system changes over time, and sensor measurements, which provide observations of the current state. It maintains a probability distribution called the *belief*, representing the estimated state and its uncertainty.
The prior belief represents the estimated probability distribution of the vehicle's state before incorporating the latest sensor measurement. It is obtained from the previous belief and the motion model.
The Bayes Filter operates through two main steps: prediction and measurement update.

A. Prediction Step

The prediction step estimates the current state using the previous belief and the vehicle's motion model. It accounts for how the vehicle is expected to move, including uncertainty in that motion.

$$\overline{bel}(x_t) = \int p\,(x_t \mid x_{t-1}, u_t) bel(x_{t-1})\, dx_{t-1}$$

where x_t is the current state, x_{t-1} is the previous state, u_t is the control input, and $p(x_t \mid x_{t-1}, u_t)$ is the motion model. The predicted belief, $\overline{bel}(x_t)$ represents the estimated state before incorporating the latest sensor measurement.

B. Measurement Update Step

The measurement update step combines the predicted belief with the latest sensor measurement to obtain a more accurate estimate of the vehicle's state.

$$bel(x_t) = \eta\, p(z_t \mid x_t)\, \overline{bel}(x_t)$$

where z_t is the sensor measurement,$p(z_t \mid x_t)$ is the measurement likelihood, and η is the normalisation constant that ensures the resulting probability distribution sums to one for discrete states or integrates to one for continuous states. The updated belief, $bel(x_t)$, represents the vehicle's state estimate after incorporating the measurement.

III. Assumptions Of The Kalman Filter

The Kalman Filter is a special case of the Bayes Filter that provides an efficient solution when certain assumptions are satisfied. These include:

1. Linear motion model: The system's state changes according to a linear relationship.
2. Linear measurement model: Sensor measurements are linearly related to the system's state.
3. Gaussian noise: Process and measurement noise follow Gaussian (normal) distributions.
4. Gaussian initial belief: The initial state estimate follows a Gaussian distribution.

Under these assumptions, the belief remains Gaussian and can be represented using only its mean and covariance, making state estimation computationally efficient.
Linear transformations of Gaussian distributions preserve their Gaussian nature, allowing the filter to update the state estimate using a finite set of equations rather than maintaining a complex probability distribution.

A. Limitations of These Assumptions

When the system is nonlinear or the noise does not follow a Gaussian distribution, the standard Kalman Filter may produce inaccurate estimates. Nonlinear systems can be handled using variants such as the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF), while Particle Filters can handle more complex, non-Gaussian distributions.

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IV. Kalman Filter

The Kalman Filter is a recursive algorithm that estimates the state of a system by combining predictions from a mathematical model with noisy sensor measurements. It consists of two main stages: prediction and measurement update. These stages are repeated as new measurements become available.

A. Prediction Step

The prediction step estimates the current state of the vehicle using its previous state estimate and control input. It also predicts the uncertainty associated with this estimate.
State prediction:

$$\hat{x}_t^- = F_t \hat{x}_{t-1} + B_t u_t$$

Where $\hat{x}_t^-$ is the predicted state, F_t is the transition matrix, $\hat{x}_{t-1}$ is the previous state estimate, B_t is the control input matrix, and u_t is the control input.
Uncertainty Prediction:

$$P_t^- = F_t P_{t-1} F_t^T + Q_t$$

Where P_t^- is the predicted covariance, P_{t-1} is the previous covariance, and Q_t represents process noise covariance.

In simple terms, the filter predicts where the vehicle will be and how uncertain that prediction is.

B. Kalman Gain

The Kalman Gain determines how much the filter should trust the predicted state compared to the new sensor measurement.

$$K_t = P_t^- H_t^T (H_t P_t^- H_t^T + R_t)^{-1}$$

Where:
K_t: Kalman Gain.
P_t^-: Predicted state covariance.
H_t: Measurement matrix.
R_t: Measurement noise covariance.
T: Matrix transpose.
(-1): Matrix inverse.

How it works:
When sensor measurements are more accurate relative to the prediction, the Kalman Gain generally increases, giving more importance to the measurement.
When sensor measurements are highly noisy, the Kalman Gain generally decreases, giving more importance to the prediction.

C. Measurement Update Step

The measurement update step corrects the predicted state using the latest sensor measurement. The Kalman Gain determines how much the measurement influences the final estimate.

State Update:

$$\hat{x}_t = \hat{x}_t^- + K_t(z_t - H_t \hat{x}_t^-)$$

where z_t is the sensor measurement, $H_t \hat{x}_t^-$ is the predicted measurement, and the difference between them is called the measurement residual.
Covariance Update:

$$P_t = (I - K_t H_t) P_t^-$$

Where, I is the identity matrix, and P_t represents the updated state covariance.
The updated estimate and covariance are then used as the starting point for the next prediction cycle.
In simple terms, the filter compares what it predicts with what the sensors actually measured, corrects its estimate, and reduces uncertainty.

V. Applications In Autonomous Vehicles

In a driverless vehicle, the Kalman Filter can be used for state estimation and sensor fusion. It combines data from sensors such as GNSS, IMUs and wheel-speed sensors to estimate the vehicle's position, velocity and orientation more accurately. Since vehicle dynamics can be nonlinear, variants such as the Extended Kalman Filter (EKF) are commonly used in practical autonomous driving systems. Within the autonomous driving pipeline, the filter processes sensor measurements and motion predictions to produce an updated estimate of the vehicle's state. This information is then used by localization, path planning and control systems to make driving decisions.

A. Role in the Driverless Pipeline

The Kalman Filter is primarily used in the state estimation stage of the autonomous driving pipeline. Sensor data from the perception and localization systems are combined with vehicle motion information to estimate the vehicle's current state. This estimate is passed to the planning and control modules, helping the vehicle determine and execute its next movement.

VI. Conclusion

The Bayes Filter provides a general framework for estimating a system's state using motion predictions and sensor measurements. The Kalman Filter simplifies this process under linear and Gaussian assumptions, offering an efficient approach to state estimation and sensor fusion. In autonomous vehicles it helps improve estimates of position, velocity and orientation, supporting reliable localization, planning and control.

References

[1] J. W. Lambert, "The Bayes Filter and Intro to State Estimation," Dec. 30, 2025. [Online]. Available: https://johnwlambert.github.io/bayes-filter/

[2] A. Becker, "Kalman Filter Explained Through Examples." [Online]. Available: https://kalmanfilter.net/